

ICE PHYSICS

Elastic ice microfibers

Peizhen Xu^{1†}, Bowen Cui^{1†}, Yejiang Bu², Hongtao Wang², Xin Guo^{1*}, Pan Wang¹, Y. Ron Shen³, Limin Tong^{1,4*}

Ice is known to be a rigid and brittle crystal that fractures when deformed. We demonstrate that ice grown as single-crystal ice microfibers (IMFs) with diameters ranging from 10 micrometers to less than 800 nanometers is highly elastic. Under cryotemperature, we could reversibly bend the IMFs up to a maximum strain of 10.9%, which approaches the theoretical elastic limit. We also observed a pressure-induced phase transition of ice from Ih to II on the compressive side of sharply bent IMFs. The high optical quality allows for low-loss optical waveguiding and whispering-gallery-mode resonance in our IMFs. The discovery of these flexible ice fibers opens opportunities for exploring ice physics and ice-related technology on micro- and nanometer scales.

Ice is one of the most abundant and important crystalline solids on Earth's surface and plays an essential role across a diverse range of topics in chemical physics, life science, geophysics, astronomy, and other disciplines (1–5). As a result, ice has been extensively studied in the past centuries (1, 2, 6–10). Bulk ice is rigid and fragile, leading to natural phenomena such as avalanches, glacier sliding, and sea ice fragmentation (1, 11, 12). The maximum elastic strain is experimentally found to be much lower than the theoretical limit of >10% (13). This difference is mostly driven by structural imperfection of real ice crystals. Materials in low-dimensional forms, such as nanoscale crystals (14–17), nanowires (18), and microfibers (19), can exhibit

far superior mechanical properties than their bulk counterparts, because of lower defect density and more uniform stress distribution (14). Nano- and microstructures of ice exist naturally in the forms of snowflakes and ice whiskers; these should be expected to also have better mechanical properties than bulk ice. Although low-dimensional ice structures such as whiskers and needles have been grown in laboratories (8, 20, 21), the focus was on growth and morphology rather than investigating mechanical properties.

We found that ice microfibers (IMFs) have exceptional mechanical properties. By adopting an electric field-enhanced growth method with a growth temperature much lower than what has previously been used (20), we suc-

ceeded in growing IMFs of high quality and small diameter (down to hundreds of nanometers). We show that the as-grown IMFs are hexagonal single crystals with the hexagonal axis along the core and have very smooth surfaces and excellent uniform cross section over their length. Our IMFs can be bent with a strain up to 10.9%, which is much higher than previously reported maximum strains (22, 23) and is a value approaching the theoretical elastic limit (14 to 16.2%) (13, 24). We conducted Raman spectroscopy measurement on bent IMFs and detected a reversible phase transition between ice Ih and II around a critical strain of 3% at -70°C . To show that the IMFs are of good optical quality, we demonstrate that they can be used to guide visible light with low optical loss and support whispering gallery modes (WGMs) around their circumference.

Growth and morphology of IMFs

The IMFs were fabricated using an electric field-enhanced growth method (20, 25) (sketched in Fig. 1A and fig. S1) (26). We grew multiple IMFs

¹State Key Laboratory of Modern Optical Instrumentation, College of Optical Science and Engineering, Zhejiang University, Hangzhou 310027, China. ²Center for X-Mechanics, Zhejiang University, Hangzhou 310027, China. ³Department of Physics, University of California, Berkeley, CA 94720, USA. ⁴Collaborative Innovation Center of Extreme Optics, Shanxi University, Taiyuan 030006, China.

*Corresponding author. Email: guoxin@zju.edu.cn (X.G.); phytong@zju.edu.cn (L.T.)

†These authors contributed equally to this work.

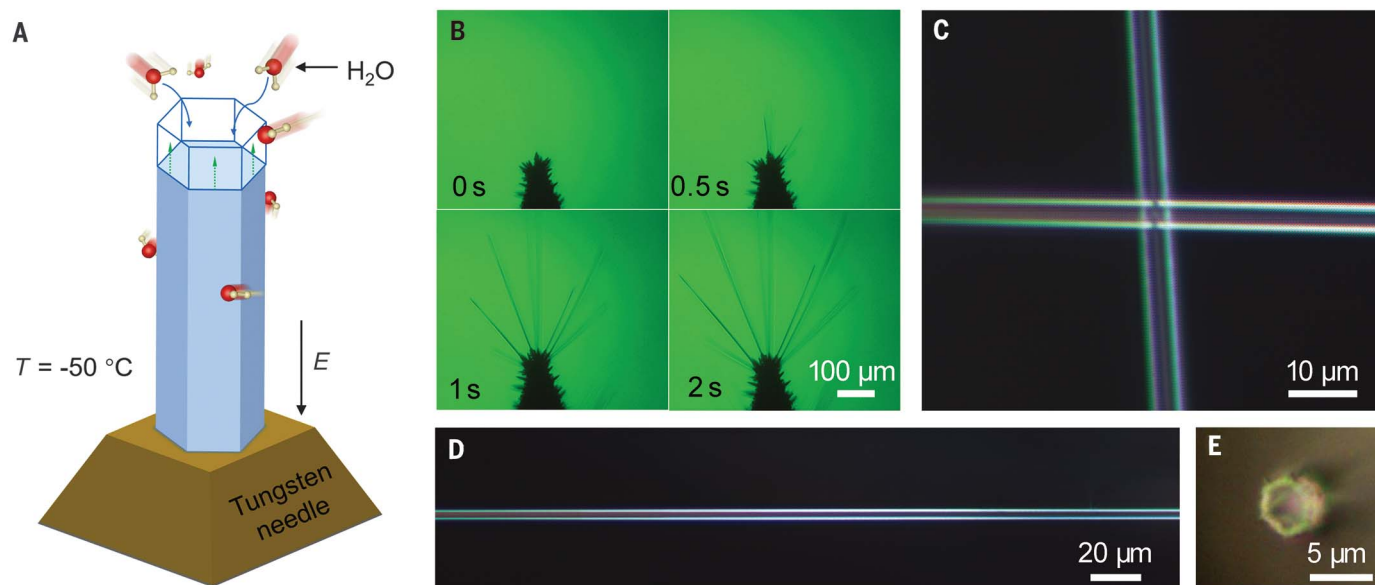


Fig. 1. Growth and optical microscopic morphology of IMFs. (A) Schematic illustration of the electrical field-enhanced growth of IMFs. A 2-kV direct-current voltage is applied to a tungsten needle in a cold chamber with a temperature (T) of -50°C . The partially shown tip of the tungsten needle serves as the base for growth of a single IMF upward along the electric field (E) direction. (B) Optical microscopic snapshots of growth in length of

multiple IMFs from the tip of a tungsten needle with a rate of $\sim 200\text{ }\mu\text{m/s}$. (C) Optical microscopic image of two crossed $3\text{-}\mu\text{m}$ -diameter IMFs, one on top of the other. (D) Optical microscopic image of a long IMF showing a uniform diameter of $\sim 3.3\text{ }\mu\text{m}$ along its length. (E) Optical microscopic image of the end face of a $4.3\text{-}\mu\text{m}$ -diameter IMF, showing a hexagonal cross section.

from the tip of a tungsten needle. We modified the method developed earlier by choosing a much lower temperature for growth [-50°C instead of -5°C as in (20); fig. S2] (26), which reduced the lateral growth rate and enabled the IMF to grow into a fiber with a smaller and more uniform diameter. We applied a 2-kV voltage to the tungsten needle to enhance diffusion of water gas molecules to the tip of the needle and to accelerate the fiber length growth. An IMF longer than $400\text{ }\mu\text{m}$ could be grown in 2 s (Fig. 1B and movie S1). After growth, we transferred the IMFs to a cold stage for characterization in different chambers (figs. S1B and S3) (26). One example we show is of two IMFs with a diameter of $\sim 3\text{ }\mu\text{m}$ forming a cross after two successive transfer operations (Fig. 1C). We also imaged a segment of a $3.3\text{-}\mu\text{m}$ -diameter IMF that illustrates the excellent lateral uniformity (Fig. 1D). Our IMFs have a hexagonal cross section (Fig. 1E), similar to what was reported earlier (20, 27).

We also used cryo-transmission electron microscopy (cryo-TEM) and cryo-focused ion

beam microscopy (cryo-FIBM) to characterize morphology and crystalline structure of IMFs (fig. S4) (26). We carried out the measurements at around -170°C . The diameter of the as-grown IMFs was typically a few micrometers (Fig. 2, A and B) but could be as small as hundreds of nanometers (Fig. 2B). A typical example of our IMFs has a tapered end with a hexagonal cross section (Fig. 2C). The root-mean-square surface roughness estimated from a high-magnification TEM image of a $2.6\text{-}\mu\text{m}$ -diameter IMF (Fig. 2D) was $<1\text{ nm}$ (fig. S5) (26), and the electron diffraction pattern (Fig. 2E, inset) reveals a $<0001>$ crystal orientation, indicating that the hexagonal c axis is aligned along the fiber length.

Elastic bending of IMFs

The bending of a fiber exerts tensile and compressive stresses on it. We used micromanipulation to bend IMFs in a cold chamber (fig. S6 and movies S2 and S3) (26). Unlike bulk ice, thin IMFs were highly flexible and could be elastically bent and readily restored to the un-

bent form. We show a set of snapshots during the bending of a $4.7\text{-}\mu\text{m}$ -diameter IMF to a minimum radius of curvature (R_c) of $63\text{ }\mu\text{m}$ (Fig. 3A, panels A1 to A4) at -70°C . When we released the bending pressure, the IMF returned to its original shape (Fig. 3A, panels A5 and A6, and movie S3). This sequence indicates a reversible elastic deformation during the bending process. To measure the strain induced during the bending of an IMF, we fit the central axis of the bent IMF with a circle and obtained the strain maxima at the inner and outer sides of the IMF (Fig. 3B) (26). Owing to the low transverse dimension and high crystal quality of the IMF, we assumed the central axis to be zero-strain and the elastic strain maxima at both sides (i.e., tensile and compressive) to be equal in absolute value. We measured IMFs of different diameters that were bent under two different temperatures, -70° and -150°C , with maximum elastic strains of $\sim 4.6\%$ (in a $4.6\text{-}\mu\text{m}$ -diameter IMF) and $\sim 10.9\%$ (in a $4.4\text{-}\mu\text{m}$ -diameter IMF), respectively; a typical set of results are shown

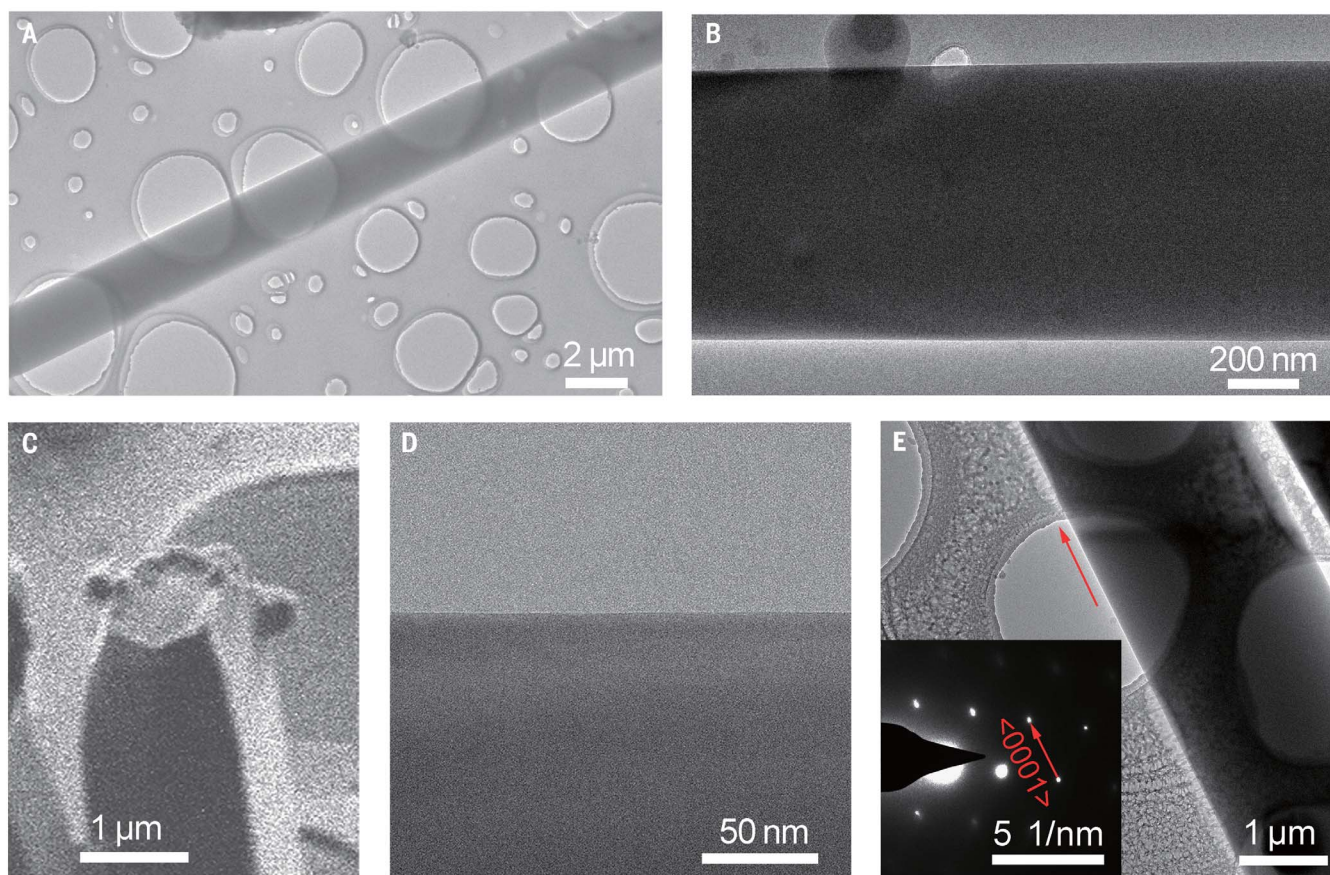


Fig. 2. Cryo-electron or ion microscopy study of IMFs. (A and B) Cryo-TEM images of IMFs with a diameter of (A) $2.1\text{ }\mu\text{m}$ and (B) 780 nm . (C) Cryo-FIBM image of a tapered end face of an as-grown IMF showing a hexagonal cross section. (D) Close-up cryo-TEM image of the surface of a $2.6\text{-}\mu\text{m}$ -diameter IMF. (E) Cryo-TEM image of a $2.3\text{-}\mu\text{m}$ -diameter IMF.

(Inset) Electron diffraction pattern revealing single hexagonal crystalline structure of the IMF with $<0001>$ orientation along the fiber. The red arrows indicate the growth direction of the IMF along the c axis. Disks and small speckles in (A) and (E) are holes on the carbon film that covers the copper grid on the TEM sample holder.

in Fig. 3C. These values are much higher than those reported in other forms of ice (e.g., <0.3%) (12, 22, 23), and the strain maxima measured at -150°C (>10%) are approaching the theoretical elastic limit (13, 24). Even having experienced ultra-large strain, all the bent IMFs returned to their original shape, indicating the absence of the type of creep that ice Ih in other forms generally suffers from under high strain (23, 28, 29). We attribute this lack of creep to the low transverse dimension and high crystal quality of IMFs and perhaps also to the relatively fast loading-unloading process we used (e.g., strain rate of $\sim 10^{-4} \text{ s}^{-1}$ for loading, and 10^3 s^{-1} for unloading; see movie S3), although we did not observe any strain rate dependence. The maximum attainable strain increased with decreasing temperature, e.g., 10.9% at -150°C versus 4.6% at -70°C (Fig. 3C), similar to the trend reported on bulk ice under

much lower strain (12) and other micro- or nanometer-scale brittle solids (14).

Ice phase transition in sharply bent IMFs

Bulk ice Ih undergoes a phase transition to ice II across a range of temperatures below -20°C under sufficiently high pressure (30, 31). At -70°C , the critical pressure is $\sim 0.2 \text{ GPa}$ (2, 23). For our bent IMF (Fig. 4A) with a maximum compressive deformation of 3%, the corresponding maximum stress calculated (using a finite element method) from the Young's modulus of ice Ih along the c axis (12.7 GPa) (32) was $\sim 0.38 \text{ GPa}$ at the inner wall of the IMF (Fig. 4B). It is therefore possible that the inner wall of the bent IMF might have already experienced the phase transition. To check this possibility, we used Raman spectroscopy (fig. S7) to probe the bent IMF (Fig. 4A) in the region where the strain is largest (26). We com-

pared the Raman spectra before and after bending (Fig. 4C) and found extra peaks at 158 and 3225 cm^{-1} in the spectrum of the bent IMF. For comparison, we included the Raman spectra for ice Ih and II (30, 31) with mode frequencies corrected by their temperature and pressure dependence (table S1) (26). The two extra peaks in the spectrum for the bent IMF coincide well with the two most prominent spectral peaks of ice II (30, 31), providing evidence of the partial presence of ice II in the bent IMF. The peaks are weak because only a small fraction of the illuminated volume of the IMF had a stress above the critical value for the Ih-to-II phase transition. The spectra were reversible on bending and unbending the IMF and were also reproducible for different IMFs with sharp bending. We observed the Raman peak of ice II within tens of seconds after bending, indicating that the

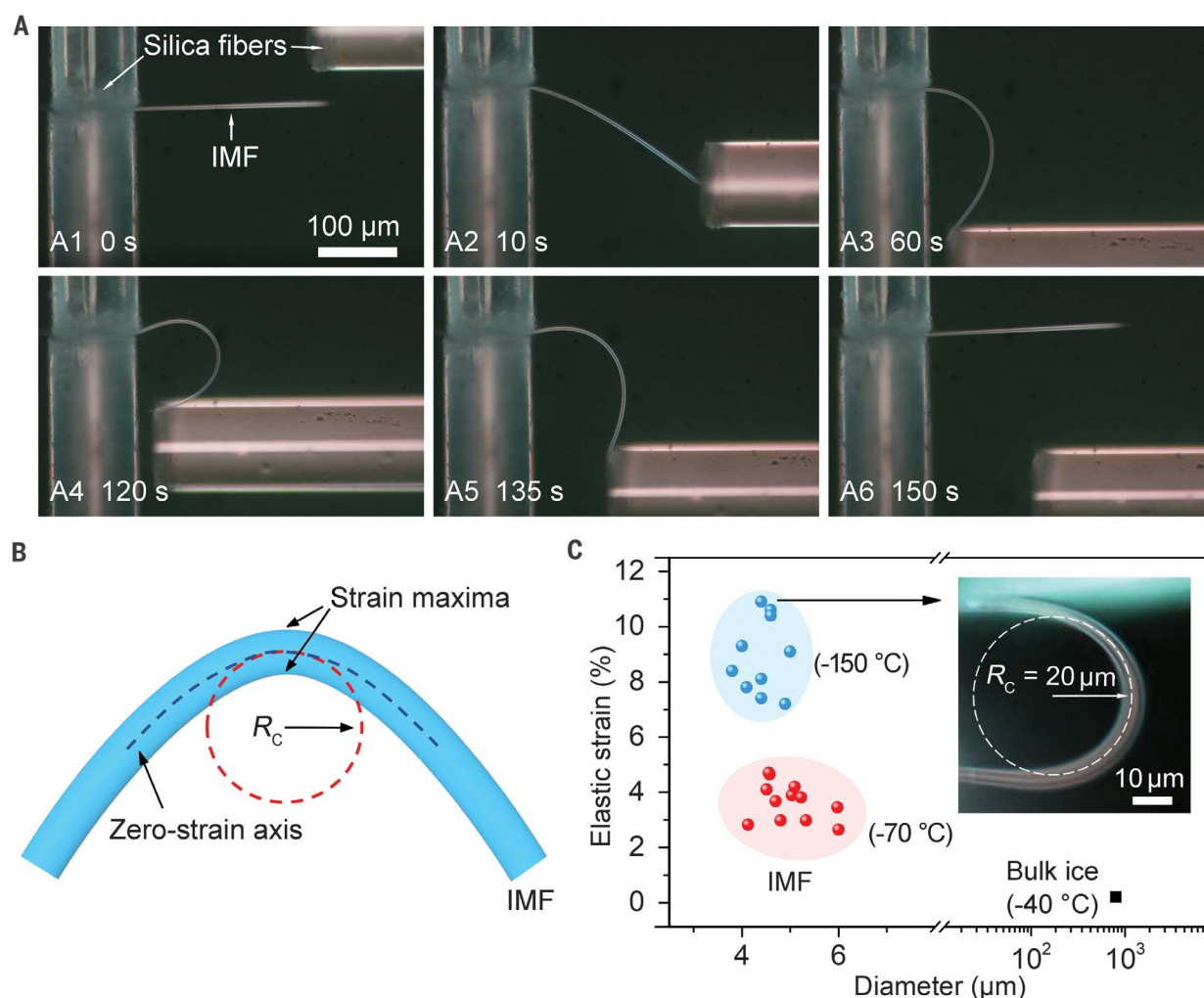


Fig. 3. Elastic bending of individual IMFs. (A) Optical microscopic snapshots illustrating how an IMF with a $4.7\text{-}\mu\text{m}$ diameter was elastically bent and unbent. (B) Schematic of the bending model of an IMF. (C) Maximum elastic strains obtained for various IMFs of different diameters bent under two different

temperatures, -70°C (red circles) and -150°C (blue circles), in comparison with that previously reported for bulk ice Ih (black square) (22, 23). The inset shows an optical microscopic image of a bent IMF ($4.4 \mu\text{m}$ in diameter and $20 \mu\text{m}$ in radius of curvature) at -150°C experiencing a maximum strain of 10.9%.

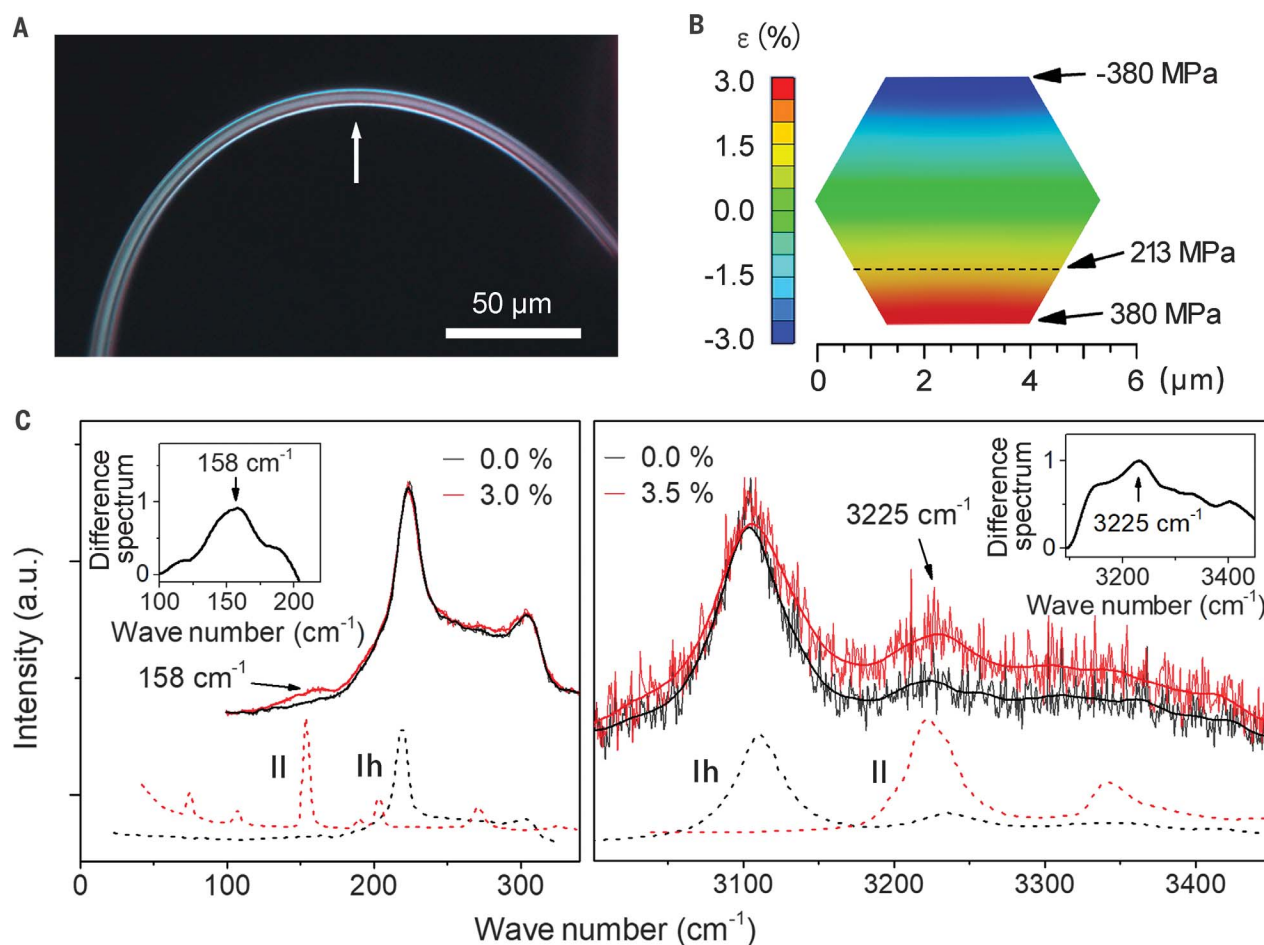


Fig. 4. Phase transition in elastic bent IMFs. (A) Optical microscopic image and (B) calculated cross-sectional strain distribution of a 5.2- μm -diameter IMF bent to a radius of curvature of $\sim 87\ \mu\text{m}$. The white arrow in (A) points at the region of maximum compressive strain. The dashed line in (B) indicates the critical pressure of phase transition for bulk ice from phase Ih to II (30, 31). (C) Raman spectra of the IMF in (A), before (black line) and after bending (red line),

obtained with probe light focused on the maximum strain region. New spectral features at ~ 158 and $3225\ \text{cm}^{-1}$, characteristic of ice II, appear in the spectrum for the bent IMF, indicating the presence of ice II. The insets show the difference spectra of the Raman spectra with and without strain around 158 and $3225\ \text{cm}^{-1}$. For reference, the Raman spectra of ice Ih (black dashed line) and II (red dashed line), adapted from (31), are shown. a.u., arbitrary units.

Ih-to-II transition of micrometer-size ice occurs rapidly, as suggested by Schulson and Fortt (33).

Optical characterization of IMFs

The excellent diameter uniformity and surface smoothness allow us to use IMFs to guide light at low temperature. By evanescently coupling visible light into one end of an IMF (Fig. 5A), we readily observed waveguiding of light of different wavelengths along the fiber length (Fig. 5B). From analysis of the position-dependent scattered light intensity along the IMF (Fig. 5C), we obtained a waveguiding loss of $\sim 0.2\ \text{dB/cm}$ at 525-nm wavelength (26), which is of the same order as those of the state-of-the-art on-chip waveguides (34). The measured loss is much higher than the absorption loss in the IMF because the absorption coefficient for ice (35) is known to be much smaller than that of typical waveguide materials such as Si_3N_4 and silica (fig. S8) (26, 36, 37). The waveguide loss we

observed must have mainly come from scattering loss, and given that surface roughness of our IMFs was at the subnanometer level, we attribute the scattering loss to structural defects in IMFs. In the bent region of an IMF, the field of the guided light is redistributed and considerably enhanced at the outer part of the bend (fig. S9). Thus, IMFs have the potential to serve as flexible waveguides with much lower loss than other optical waveguides within the visible spectral range and to be used for optical applications such as high-sensitivity microfiber-based optical sensing at low temperature (38). To explore the possibility of exciting WGMs on the circumference of IMFs, we focused a white light beam perpendicularly on an IMF and collected the scattered signal with an objective in the dark-field configuration (fig. S10) (26). We found a typical spectrum of a transverse electric (TE) WGM with a quality factor of ~ 60 and a free spectral range of $\sim 20.9\ \text{nm}$

at around 605-nm wavelength, which agrees well with the calculated value ($\sim 22.0\ \text{nm}$) of the TE_{26} WGM in the 4.4- μm -diameter IMF (26) (Fig. 5D). The WGM has enhanced field intensity on the IMF surface (Fig. 5E), making it also sensitive to surface or environmental changes.

Summary and prospects

We have succeeded in fabricating high-quality single-crystalline IMFs with a diameter down to a few hundreds of nanometers and demonstrated that they are highly flexible and can be reversibly deformed by bending to a radius of a few tens of micrometers. We obtained a lattice compression (tension) of 10.9% on the inner (outer) side of a bent IMF, which is more than one order of magnitude larger than those reported in other forms of ice and approaches the theoretical elastic limit (13, 24). In sharply bent IMFs, we observed phase transition between ice Ih and II when the compressive strain

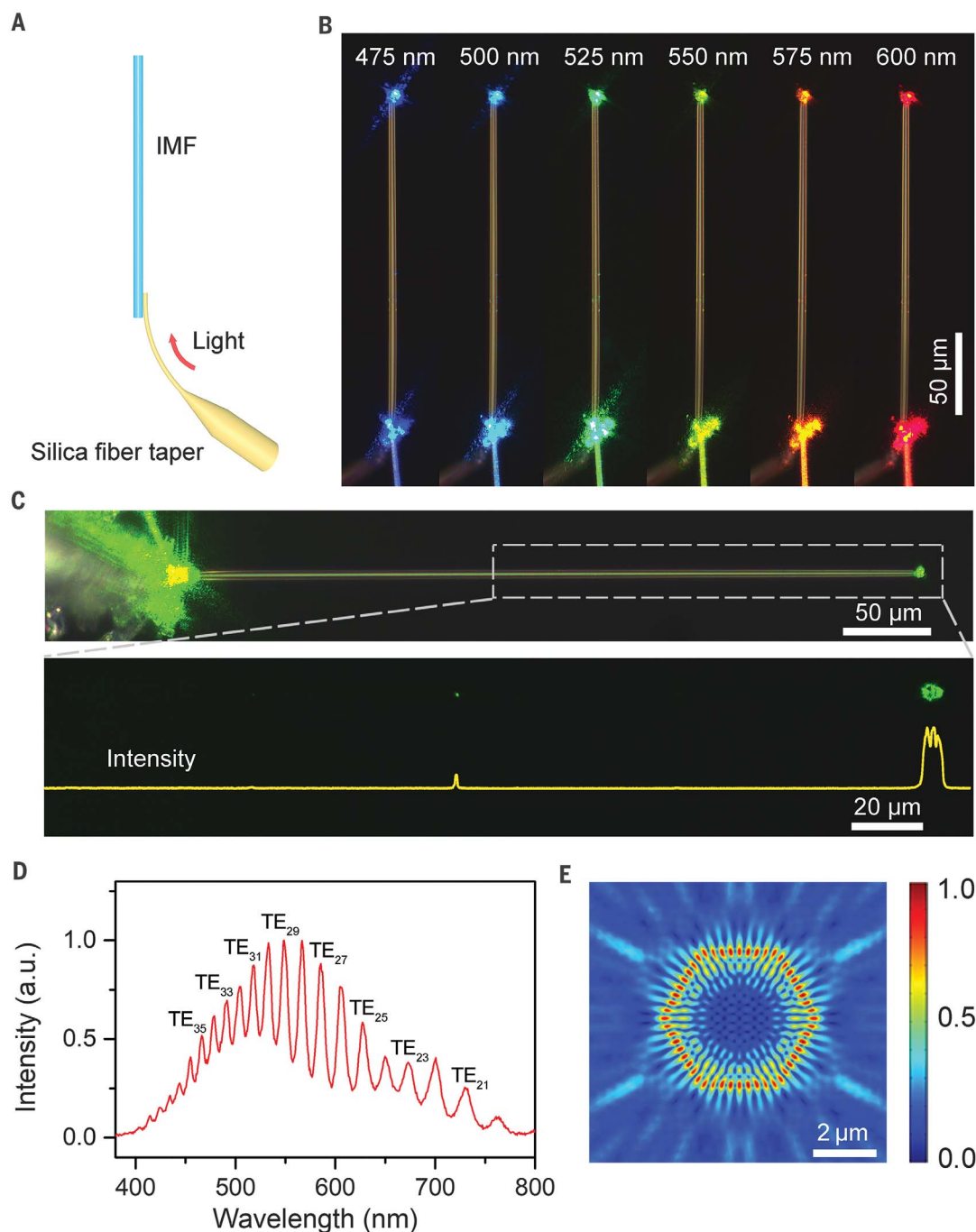


Fig. 5. Optical characterization of IMFs. (A) Schematic diagram of launching light into an IMF by the evanescent wave coupling method. (B) Microscopic images of an IMF (4.4 μm in diameter and 200 μm in length) guiding light of different wavelengths. (C) (Top) Bright-field microscopic image of a 5.4-μm-diameter IMF guiding 525-nm light. The section included in the dashed box was

used in collecting scattered light. (Bottom) Dark-field microscopic image of the boxed section of the IMF in the top frame, showing very weak intensity of scattered light from the IMF except for a spot near the center. (D) Spectrum of TE WGM of a 4.4-μm-diameter IMF detected from light scattering from the IMF. (E) Simulated electric field distribution of the TE₂₆ WGM on the IMF in (D).

exceeded 3%, suggesting a simple means to study phase transitions of ice using the bending of IMFs. Higher strains are achievable with sharper bending of IMFs at lower temperature, allowing study of ice-phase transitions of Ih not only to II but also to phases III, V, VI, and IX (10). Furthermore, the outer side of a bent IMF

should experience a tensile stress (negative pressure) of equal magnitude to the compressive stress on the inner side and can also be used to study ice phases and transition dynamics under large negative pressure (39, 40), which are known to be experimentally challenging (41). We have also demonstrated that

IMFs can serve as optical waveguides with very low loss and support WGMs in the visible spectrum, both of which are means to enhance light interaction with matter. We could imagine the use of IMFs as low-temperature sensors to study, for example, molecular adsorption on ice, environmental changes, structural variation,

and surface deformation of ice [e.g., surface premelting (42)]. In short, the elastic IMFs demonstrated here may offer an alternative platform for exploring ice physics and open previously unexplored opportunities for ice-related technology in various disciplines.

REFERENCES AND NOTES

- V. F. Petrenko, R. W. Whitworth, *Physics of Ice* (Oxford Univ. Press, 2002).
- T. Bartels-Rausch et al., *Rev. Mod. Phys.* **84**, 885–944 (2012).
- C. Hoose, O. Möhler, *Atmos. Chem. Phys.* **12**, 9817–9854 (2012).
- R. Staroszczyk, *Ice Mechanics for Geophysical and Civil Engineering Applications* (Springer Nature, 2019).
- L. O'Rourke et al., *Nature* **586**, 697–701 (2020).
- J. Kepler, *The Six-Cornered Snowflake* (Oxford Clarendon Press, 1966).
- P. W. Bridgman, *Proc. Am. Acad. Arts Sci.* **47**, 441–558 (1912).
- T. W. Zawidzki, H. M. Papee, *Nature* **196**, 568–569 (1962).
- G. Malenkov, *J. Phys. Condens. Matter* **21**, 283101 (2009).
- C. G. Salzmann, *J. Chem. Phys.* **150**, 060901 (2019).
- N. Urabe, T. Iwasaki, A. Yoshitake, *Cold Reg. Sci. Technol.* **3**, 29–37 (1980).
- J. J. Petrovic, *J. Mater. Sci.* **38**, 1–6 (2003).
- P. A. Santos-Flórez, C. J. Ruestes, M. de Koning, *J. Chem. Phys.* **149**, 164711 (2018).
- T. Zhu, J. Li, *Prog. Mater. Sci.* **55**, 710–757 (2010).
- A. Banerjee et al., *Science* **360**, 300–302 (2018).
- A. Nie et al., *Nat. Commun.* **10**, 5533 (2019).
- C. Dang et al., *Science* **371**, 76–78 (2021).
- G. Brambilla, D. N. Payne, *Nano Lett.* **9**, 831–835 (2009).
- L. Tong et al., *Nature* **426**, 816–819 (2003).
- K. G. Libbrecht, T. Crosby, M. Swanson, *J. Cryst. Growth* **240**, 241–254 (2002).
- R. Ma et al., *Nature* **577**, 60–63 (2020).
- E. M. Schulson, *Eng. Fract. Mech.* **68**, 1839–1887 (2001).
- E. M. Schulson, P. Duval, *Creep and Fracture of Ice* (Cambridge Univ. Press, 2009).
- Z. M. Jendi, P. Servio, A. D. Rey, *Cryst. Growth Des.* **15**, 5301–5309 (2015).
- K. G. Libbrecht, V. M. Tanusheva, *Phys. Rev. Lett.* **81**, 176–179 (1998).
- Materials and methods are available as supplementary materials.
- K. G. Libbrecht, *Annu. Rev. Mater. Res.* **47**, 271–295 (2017).
- M. Mellor, D. M. Cole, *Cold Reg. Sci. Technol.* **6**, 207–230 (1983).
- J. Weertman, *Annu. Rev. Earth Planet. Sci.* **11**, 215–240 (1983).
- B. Minceva-Sukarova, W. F. Sherman, G. R. Wilkinson, *J. Phys. C Solid State Phys.* **17**, 5833–5850 (1984).
- A. K. Garg, *Phys. Status Solidi A* **110**, 467–480 (1988).
- J. J. Neumeier, *J. Phys. Chem. Ref. Data* **47**, 033101 (2018).
- E. M. Schulson, A. L. Fortt, *J. Geophys. Res. Solid Earth* **117**, B12204 (2012).
- T. Horikawa, D. Shimura, T. Mogami, *MRS Commun.* **6**, 9–15 (2016).
- S. G. Warren, R. E. Brandt, *J. Geophys. Res. Atmos.* **113**, D14220 (2008).
- H. R. Philipp, *J. Electrochem. Soc.* **120**, 295–300 (1973).
- R. Kitamura, L. Pilon, M. Jonasz, *Appl. Opt.* **46**, 8118–8133 (2007).
- J. Lou, Y. Wang, L. Tong, *Sensors* **14**, 5823–5844 (2014).
- Y. Huang et al., *Sci. Adv.* **2**, e1501010 (2016).
- T. Matsui, T. Yagasaki, M. Matsumoto, H. Tanaka, *J. Chem. Phys.* **150**, 041102 (2019).
- A. Falenty, T. C. Hansen, W. F. Kuhs, *Nature* **516**, 231–233 (2014).
- Y. M. Li, G. A. Somorjai, *J. Phys. Chem. C* **111**, 9631–9637 (2007).

ACKNOWLEDGMENTS

We thank X. K. Zhang, L. Y. Wu, and S. H. Chang at the Center of Cryo-Electron Microscopy (CCEM), Zhejiang University, for their technical assistance on cryo-TEM; X. J. Wang at the cryo-EM facility of Westlake University for the technical assistance on cryo-FIB/SEM; and L. Yang and Y. Y. Jin for assistance with scattering spectra analysis. This work was supported by the National Key Research and Development Project of China (2018YFB2200404), the National Natural Science Foundation of China (11527901), the Natural Science Foundation of Zhejiang Province (LR21F050002), and Fundamental Research Funds for the Central Universities. **Author contributions:** L.T., X.G., and Y.R.S. conceived of and supervised the project. P.X. and B.C. performed the experiments. Y.B., P.X., and B.C. performed the cryo-TEM measurements. All authors analyzed the data. L.T., X.G., P.X., B.C., and Y.R.S. wrote the manuscript. All authors discussed the results and commented on the paper. **Competing interests:** The authors declare no competing interests. **Data and materials availability:** All data are available in the main text or the supplementary materials.

SUPPLEMENTARY MATERIALS

science.sciencemag.org/content/373/6551/187/suppl/DC1

Materials and Methods

Figs. S1 to S10

Table S1

References (43–46)

Movies S1 to S3

6 March 2021; accepted 2 June 2021

10.1126/science.abh3754



Elastic ice microfibers

Peizhen Xu, Bowen Cui, Yejiang Bu, Hongtao Wang, Xin Guo, Pan Wang, Y. Ron Shen, and Limin Tong

Science **373** (6551), . DOI: 10.1126/science.abh3754

Ice goes bendy

One well known characteristic of ice is that it fractures instead of bending when strained. This characteristic is caused by the inevitable defects introduced into the ice structure during solidification. Xu *et al.* show that very thin, carefully grown ice microfibers can bend a lot, up to about 11%, and still remain elastic (see the Perspective by Schulson). This value is reasonably close to theoretical limits previously estimated. The fibers are also super clear, allowing for efficient light transmission.

Science, abh3754, this issue p. 187; see also abj4441, p. 158

View the article online

<https://www.science.org/doi/10.1126/science.abh3754>

Permissions

<https://www.science.org/help/reprints-and-permissions>

Use of this article is subject to the [Terms of service](#)

Science (ISSN 1095-9203) is published by the American Association for the Advancement of Science. 1200 New York Avenue NW, Washington, DC 20005. The title *Science* is a registered trademark of AAAS.

Copyright © 2021 The Authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original U.S. Government Works